

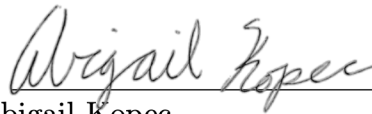
UTILIZING COMPUTER MODELING TO OPTIMIZE
ELECTRIC FIELDS WITHIN XENON TIME
PROJECTION CHAMBERS

by

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A Thesis Submitted to the Honors Council
For Honors in the Department of Computer Science
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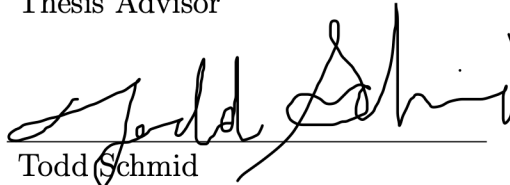
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Abstract

XENONnT is a physics experiment designed with the goal of detecting dark matter particles. The detector is a time projection chamber; a series of charged electrodes creates an electric field, surrounding a central body filled with liquid and gaseous xenon. Photomultiplier tubes (PMTs), positioned on either end of the chamber, serve to detect light signals. In looking to optimize the detector, we seek to minimize the root mean square of the electric field norms experienced by the PMTs. This quantity corresponds to the variance in the electric field observed by the PMTs. Establishing a consistent electric field is important to maintaining these sensitive components. The electric field's distribution can be modulated via two screening rings positioned near the PMTs, one on either end of the detector. Using the simulation software COMSOL we adapted a pre-existing model of the detector to evaluate this quantity for different simulation parameters. We determined the ideal voltage for each of these screening rings based on the voltage of the other components within the chamber, finding a linear relationship between the voltage of the Anode and Cathode to the ideal voltage of the screening rings. We also determined the average electric field magnitude on the PMTs for a specific Anode or Cathode voltage. Additional factors

were also evaluated, including a comparison of the simulated XENONnT in its current state to a past iteration of the detector where it possessed wire meshes in place of the screening rings. While the meshes lead to lower overall field magnitudes on the PMTs, the performance from preliminary tests suggests that the conditions for the PMTs with the screening ring are acceptable.

Chapter 1

Introduction

In our search to understand the nature of the universe, particle interactions provide key insights into the fundamental mechanics that inform everyday physical phenomena. A time projection chamber (TPC) is a specialized particle detector that can be used to create a three-dimensional reconstruction of particle interactions. These two-dimensional detector arrays can extrapolate three-dimensional position via temporal data to extract depth information. The XENONnT time projection chamber [3], seen in figure 1.2, is a TPC that uses 5.9 tons of liquefied xenon as the particle interaction target. The detector has the goal of identifying WIMPs, hypothesized particles that are one of the primary candidates for dark matter [4].

Both the top and bottom of the detector are lined with a 2-D array of photomultiplier tubes (PMTs) that detect photons. As seen in figure 1.1, when a particle interaction occurs, the liquid xenon contained within the TPC is excited, releasing

light and electrons. This burst of light is detected by the PMTs. The electrons are then brought upwards in a uniform electric field created by an Anode and Cathode mesh located at opposite ends of the chamber. Just below the Anode, they pass through a Gate that modifies electric field strength, with the field being more intense above the Gate. Accelerating after passing through the Gate, the electrons induce fluorescence in gaseous xenon, creating a second flash of light. The location of this burst is used to determine the distance on the x and y axes. The temporal distance between the initial burst of light caused by the collision and the secondary burst is used to determine depth on the z -axis, since an event near the top of the chamber will take less time to reach the Anode and create the fluorescent burst. This determination of depth is contingent upon the electric field being as uniform as possible.

The construction of XENONnT involves a series of charged surfaces and electrode meshes surrounding the active liquid volume of the experiment. At the top of the chamber, positioned below the PMTs within the gaseous xenon, is a screening ring, known as the Top Screening Ring (TSR). The TSR serves to protect the top PMTs from ions and strong electric fields. Below the TSR are the Anode and Gate. These components, listed in table 1.1 from top to bottom, are detailed in figure 1.3. Surrounding the body of the liquid xenon is a field cage, a series of 64 shaping rings that encircle the vast majority of the active liquid volume and are connected by resistors that act to regulate the electric fields within the model [2]. Below this is a Cathode which acts to repel the electrons excited by interactions, followed finally by a second screening ring, the Bottom Screening Ring (BSR). The BSR protects the bottom PMTs from strong electric fields caused by the Cathode. These bottom components

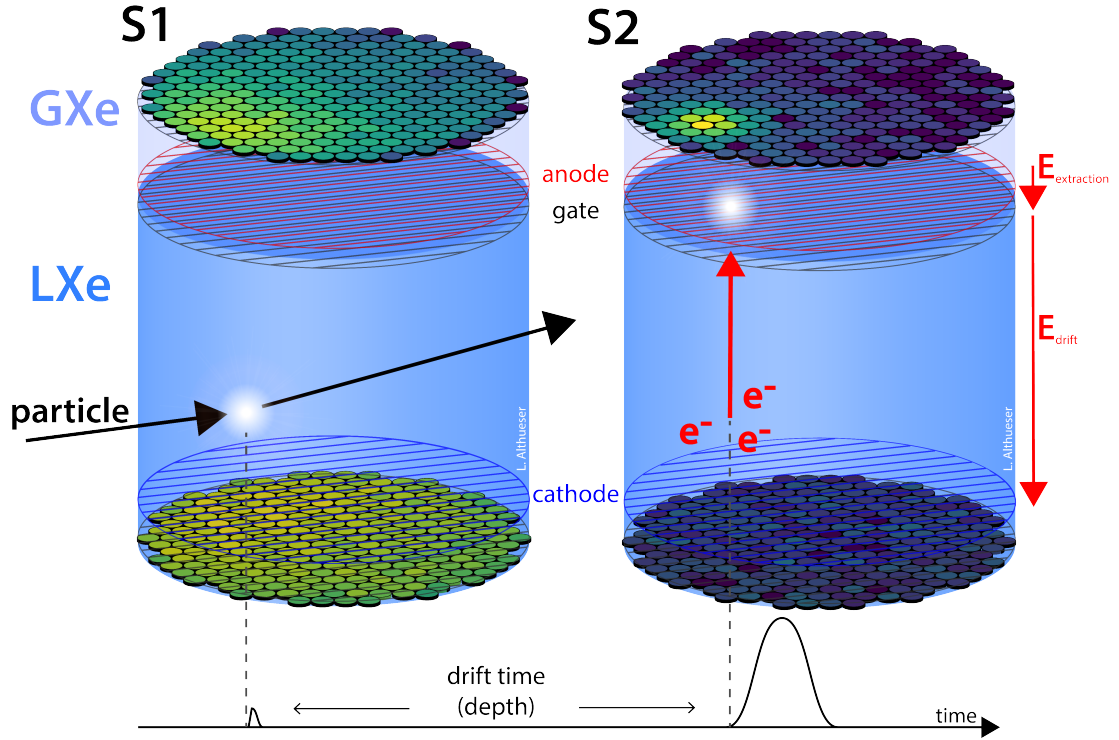


Figure 1.1: A diagram of the principles of a Time Projection Chamber. First, an interaction produces a burst of light. After some time, a second localized burst of light is seen when electrons, transported through the fluid body, collide with the Anode. Depth is calculated via the time between these interactions. Credit: XENON Collaboration

are shown in figure 1.4, which extends from the field shaping rings to the bottom of the chamber.

In order to accomplish its goal of maximum sensitivity to interactions, XENONnT seeks to optimize several key properties. Two notable properties have already been optimized via the field shaping rings. First among these properties is the charge-insensitive mass, a unit describing the mass of the xenon within the detector body that is insensitive to particle interactions. These regions are insensitive as the elec-

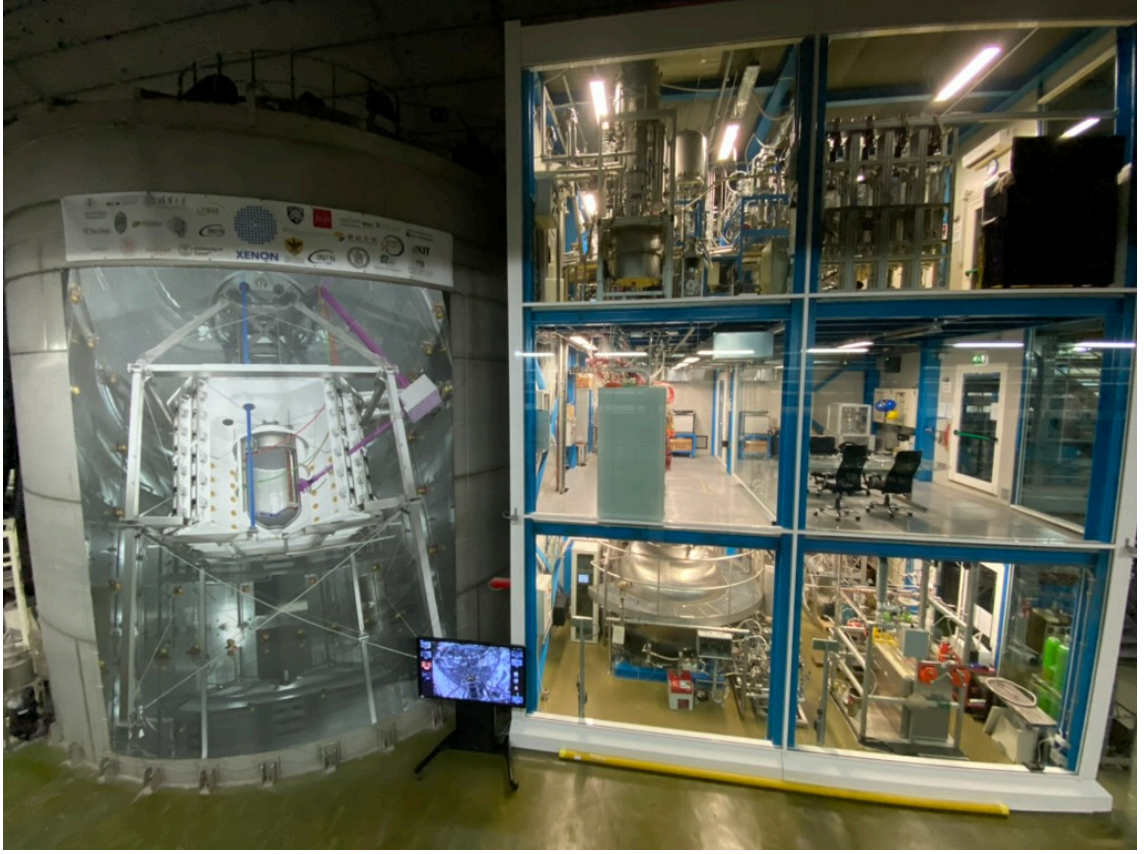


Figure 1.2: Outer view of the XENONnT experiment at LNGS. Credit: XENON Collaboration

tric field lines in these zones do not send electrons to the Anode, but instead to collisions with the walls. Second is the field spread, or maximal variance between field strength within the charge-sensitive region of the liquid xenon. This variance must be minimized as it introduces variance to the delay between interaction and detection, compromising experimental results and adding increased risk of error to measurements. XENONnT has worked to minimize charge-insensitive mass and field disparities via adjustments to the field shaping rings [2]. Values are optimized via detector hardware (specifically, the resistance between field shaping rings), which is

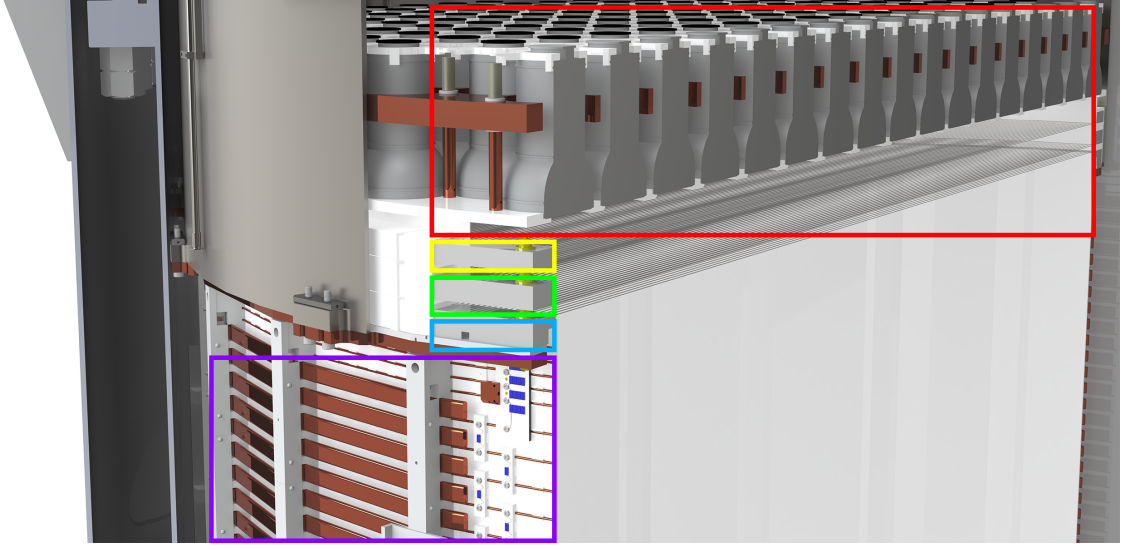


Figure 1.3: Detailed illustration of mechanics at top of XENONnT, showing the PMT array (red) and the three top meshes: from top to bottom, the TSR (yellow), Anode (green), and Gate (cyan). Below these are the field shaping rings, made of copper (purple). Credit: XENON Collaboration

a fixed constraint at this point in time due to the completion of the detector.

A third quantity that can be optimized is the electric field variance experienced by the PMTs. Voltage, represented V , is a measurement of Electrical Potential. Electric Field is defined as the gradient of the Electrical Potential [6] via the equation $E = -\nabla V$. Electric field norm is the magnitude of this gradient without direction, $E_n = |E| = |\nabla V|$. The primary scope of this report is the RMS, or root mean square, of the electric field magnitude on the surface of the PMTs. This is an optimal way to quantify the distribution of the electric field on the surface of the PMTs, with the RMS allowing us to measure the amplitude of variations. A small RMS implies a uniform distribution of electric field norms. Ensuring a small RMS will allow us to ensure the safety of PMTs; a non uniform electric field might cause strain on the

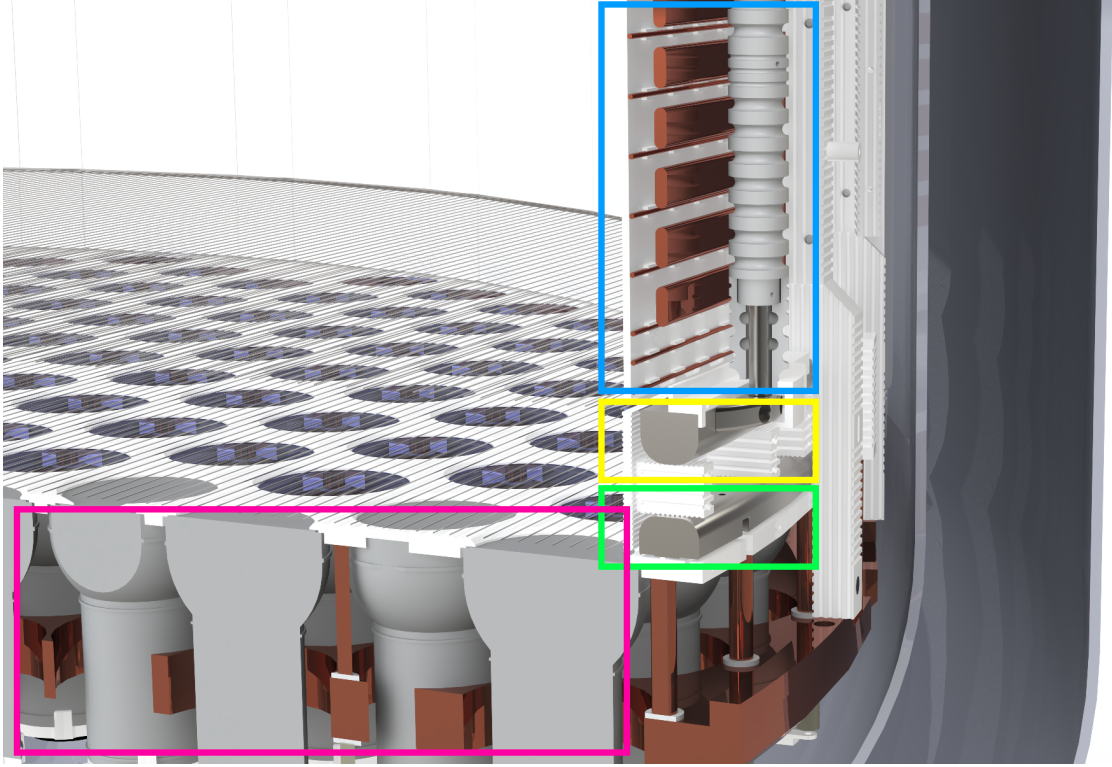


Figure 1.4: Detailed illustration of mechanics at bottom of XENONnT, showing from top to bottom: field shaping rings (blue), the Cathode (yellow), BSR (green), and PMT array (magenta). Credit: XENON Collaboration

PMTs and damage them. Using the TSR and BSR we can adjust the electric fields around the edge of the XENONnT chamber, which on both the top and bottom PMT arrays sees the greatest variance in electric field strength. Determining the correct voltages for these screening rings is therefore vital in ensuring the safety of the detector's PMTs.

However, conducting detailed configuration tests on XENONnT to determine the RMS would be impractical due to constraints on time, location, and safety. Working with XENONnT is time consuming, and if values are miscalibrated the system could be heavily damaged, requiring months of costly repairs. Furthermore, conducting

Component	Minimum Voltage	Maximum Voltage
Upper PMT Array	-1.3 kV	-1.3 kV
TSR/Mesh	TBD	TBD
Anode	0 kV	8 kV
Gate	-8 kV	0 kV
Top Field Shaping Ring	0 kV	0 kV
Cathode	-20 kV	-10 kV
BSR/Mesh	TBD	TBD
Lower PMT Array	-1.3 kV	-1.3 kV

Table 1.1: The voltage ranges for the charged components within XENONnT, with components of invariable voltage (whose values are not in the scope of this study) being listed with identical minimal and maximum voltage.

these tests on the physical XENONnT would require the installation of electric field sensors which do not already exist. Thus, simulation provides the most effective way to analyze the optimizations of the screening rings.

To conduct the complex simulations of the XENONnT system, the multiphysics software COMSOL is utilized [5]. COMSOL is a finite element simulation system, meaning it divides a space into a mesh of individual vertices at variable resolution, whose properties are determined by the materials they represent. After analyzing the subproblems for each individual segment of the mesh, the system is then recombined in order to determine the final state of the system. This allows a far greater degree of local accuracy than a global system of equations would, and allows the incorporation of materials with variable properties.

In our use of COMSOL for simulations we will assess various combinations of voltage for the Anode and Cathode and their corresponding screening rings, determining

their effect on the PMT electric field distribution. From here we aim to determine the RMS for each combination, and from there determine the relationship between Anode and Cathode voltage and the ideal voltage of the screening rings. This will determine a precise voltage to set the screening rings to for any chosen value of the Anode and Cathode within our scope, ensuring the safety of the PMTs.

Chapter 2

Methodology Utilizing COMSOL

The COMSOL workflow makes up the majority of the time spent gathering and processing data for this analysis. COMSOL uses a finite element workflow, which divides a space into a mesh of individual vertices at variable resolution. The properties of the mesh are determined by the materials they represent, and each node can calculate its values based on its neighbors. COMSOL defines geometries and gathers results using the following steps, which we break up into the following sections:

Global Definitions. Global Definitions is the first COMSOL section. These are parameters established at the beginning of a workflow. These definitions can correspond to parameters in physical space, such as the width of a PMT, or to other values, such as the voltage of a PMT. Defining these parameters allows for a more adaptable construction of models, and a greater ability to adjust parameters between experimental evaluations. This section contains a reference

table to all defined materials by default.

Definitions. This is the first section contained within a specific component used for study. It has a similar function to the Global Definitions, but where the scope of Global Definitions extends to all components (independently created geometries), Definitions only extends to a single component. Only one component is used in the following studies, rendering the difference between this section and the prior one relatively minor.

Geometry. This section defines the physical geometry of the component, creating sectioned off areas mathematically with parameters from Definitions and Global Definitions. Different areas of the geometry can later be assigned different material properties. Not all geometrically defined areas must correspond to physical properties; lines or areas that one wants to evaluate data on in later parts of the study can also be defined here, though the creation of a nonphysical line of evaluation can also be performed within the Results section.

Materials. This section determines which materials defined in Global Definitions are used in the model. Areas of the Geometry defined prior are assigned the properties of a chosen material. One material can also be designated the default material, with unassigned areas given the properties of said material. The default material in this study was liquid xenon.

Multiphysics Models. This is the phase of the workflow where properties relating to physics modules are incorporated. COMSOL, as a multiphysics modeling software, has a large number of physics simulations such as fluid simulations, mechanical stress simulations, and electrical current simulations, which are not

within the scope of most studies. For this reason, each of these modules can be added to a simulation independently. In the case of simulations for XENONnT performed in the course of the study, only the Electrostatics Simulation is necessary. In this model, edges and areas are given voltages from the values defined in Global Parameters.

Mesh. A mesh, within the scope of finite element simulations, is a means of translating over the vectorized, infinite-resolution geometry into a graph which corresponds to it. Each vertex in the mesh corresponds to a point in the Geometry, taking its defined material properties. Areas with high geometric complexity can be given more high-density meshes to ensure that detailed parts of the simulation, such as corners or edges, are given more attention than empty spaces. The Mesh section defines the mesh used within the model, giving the user control over the resolution of the mesh in different sections of the model.

Study. This step allows the user to evaluate the model defined in the earlier steps. It gives the user two options for evaluation: **Compute**, which is slower and runs the model from the beginning, and **Update Solution**, which is faster but does not re-evaluate parameters. This section also gives the user the choice to define plots by default, the choice to save the data from the model, and other quality of life features.

Results. The results section allows the user to analyze the results of the study evaluated prior. It contains the datasets generated from the study, including subsets of the broader dataset which can be generated after the completion of the study (such as zones or lines of evaluation). As well, this section possesses means of

interfacing with datasets. These include 1D and 2D graphs and charts, as well as the ability to export data. This export ability is the most critical use of the results section within this analysis.

This study used COMSOL version 6.3 on a Linux machine [5]. To use COMSOL on a machine where it is downloaded, the following commands are used:

```
$ module load comsol
$ comsol
```

In order to conduct our analysis, a model previously created by Dr. Francesco Toschi [9] and Dr. Vera Hiu-Sze Wu [10] for XENONnT was used. This model is axisymmetric, meaning it is a 2D model evaluated as a 3D system. One axis of the 2D model represents radial distance, with another representing vertical distance. These two dimensions are labeled r and z respectively. The model is then revolved around the axis to create a 3D space. This gave us the basis for our model's definitions, geometry, materials, and electrostatics model, as well as the configuration scheme for the mesh. However, changes had to be made to the model's geometry and assignments before use. The mesh resolution and configuration did not need to be adjusted because the existing file had a successful mesh implementation that adapted to changes in the geometry.

The most significant geometric change to the model is the creation of quartz windows within the PMTs: the PMTs possess these protective insets on their frontward faces, which measure 3.5 mm thick. In the original model, the line defining the surface of the PMT was given a fixed voltage of -1.3kV; after the addition of the quartz

windows, this line was moved to the internal face of the quartz windows. This better represents the location of the charged photocathode that converts light into electrons to be read out as a current [1]. While the intention was to assign the quartz windows the quartz material, a crash in COMSOL reset their material properties to that which was originally assigned to the regions (liquid/gaseous xenon, based on whether they were above/ below the liquid surface); this discrepancy was discovered after the data collection phase of this project was already complete. As seen in figure 2.1, this error does not significantly affect the conclusions of this work. Important future work would be to re-run all simulations, now knowing the extent to which COMSOL shutting down removes the memory of material processes.

Electric Field along surface of Cathode PMTs based on model material of Quartz Windows

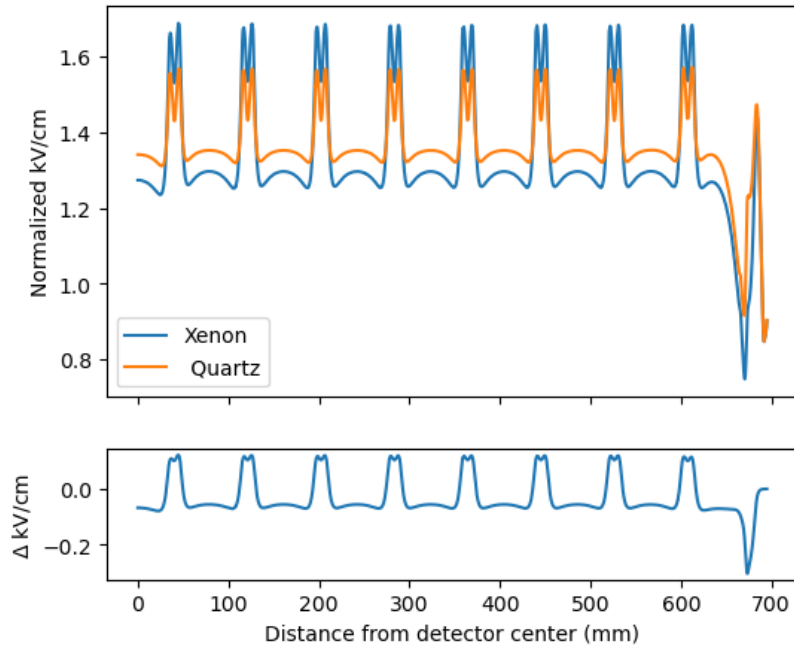


Figure 2.1: The comparison between Electric Field Norms on the bottom PMTs with a Cathode at -10kV and a BSR at -2kV. Note that the variance over the central PMTs, which are responsible for the mean Electric Field Norm, have a low Delta of approximately 0.05 kV/cm.

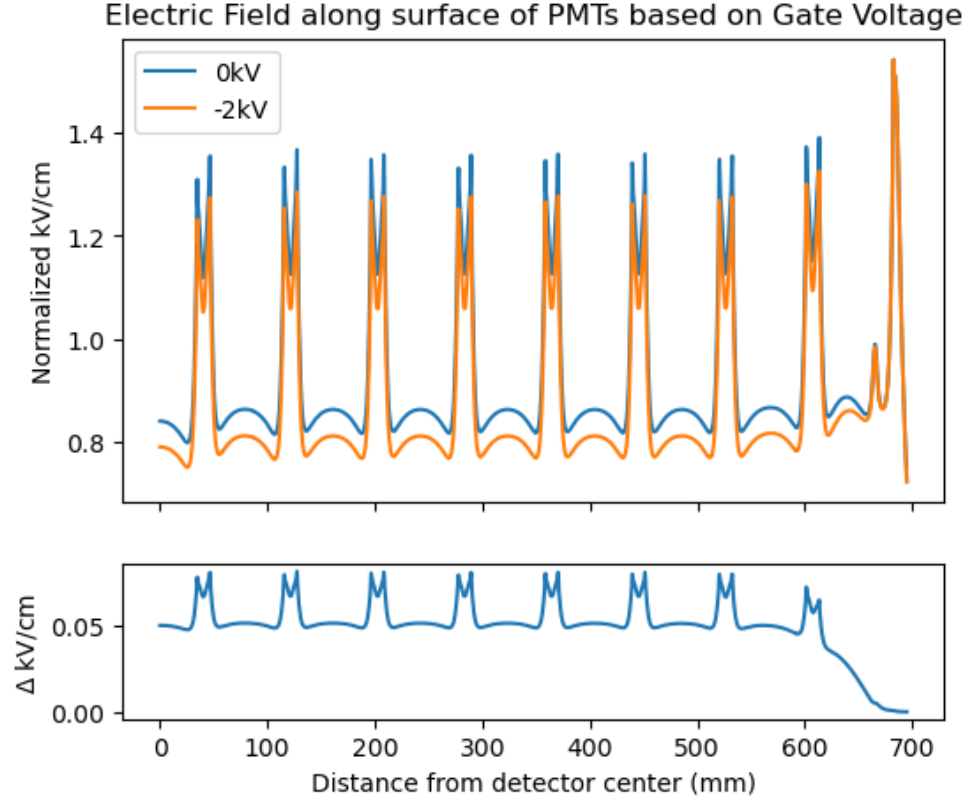


Figure 2.2: A comparison of the results of two data runs assuming the same Anode and TSR voltages (6kV and 0.4kV respectively) with different gate voltages. Note that the actual conditions experienced by the critical outer PMT do not differ as greatly as those experienced by the other PMTs. Even though the outer PMT is minimally affected, the change to the electric field norm experienced by the inner PMTs would change the mean, and thus the RMS.

Originally the XENONnT experiment was intended to have two screening meshes in place of the screening rings seen now. These were connected to the top and bottom screening rings (TSR and BSR), and were meant to be Faraday Cages [6] that dissipated high electric fields away from the PMTs. A broken wire on the bottom screening mesh led to a failure of the Cathode early in XENONnT. As a result it was decided in Summer 2025 that the wires should be removed and replaced with rings.

This removed the risk of broken wires causing damage to sensitive components in the future. However, this new setup leaves the PMTs exposed to fields from the Cathode and Anode.

The COMSOL geometry originating from Toschi and Hiu-Sze Wu’s study, made before this change, possessed these screening meshes. These wires had to be removed to align the model with current desires for the detector. COMSOL constructs its geometries through a timeline of events that can be individually deactivated; the wires were thus removed in this manner, deactivating their construction while keeping their geometry intact for future use of the model. As well, several minor changes were made to the buffers at the outer edge of the bottom PMTs, with others likewise performed to the top PMTs. These changes extended buffers made of Poly-tetrafluoroethylene (PTFE) – more commonly known by its brand name Teflon – to align them with parameters displayed in blueprints for the detector.

After the updates to the geometry were performed, the COMSOL finite element mesh (not to be confused with the screening meshes) was generated. The way of generating the mesh was from Vera and was deemed effective [10]. There was no need to manually adjust the mesh.

The next phase of the COMSOL workflow was updating parameters. As the physical geometry was not changing, all parameters that changed between studies related to voltages. These are in a pre-made parameter set, `XENONnT_potential`, located in the global definitions. Relevant to us are six parameters, each of which controls a voltage in the model:

TopScreenPotential: controls potential of TSR.

AnodePotential: controls potential of Anode.

GatePotential: controls potential of Gate.

CathodePotential: controls potential of Cathode.

BottomScreenPotential: controls potential of BSR.

PMTPotential: controls potential of PMTs. Should always be set to -1.3 kV.

These values are set according to the parameters that one wishes to evaluate for a specific simulation. It was assumed in this study that non-local contributions to electric field norms are negligible; i.e., the voltage of the Anode will not have an effect on the electric field norm experienced near the Cathode. In general, it was assumed that the electric field norm experienced by the PMTs would only depend on the nearest electrode (Anode/Cathode) and the nearest screening ring (TSR/BSR). The exception to this rule is that the close proximity of the Anode and Gate led to the Gate having a minor but noticeable effect on the top PMTs.

After values for the parameters are chosen in accordance with the values desired, the simulation is run using the Study section. Specifically, the **Compute** option is chosen over **Update Solution** as the latter does not pass through changes made to the parameters. Once the simulation has run (taking 5-10 minutes on our machine) one can proceed to the Results section. COMSOL automatically brings the user to the Results window once a simulation has finished running.

In order to analyze the results of a completed study, one must first define a line of evaluation. This line is a “slice” of the 2D results of the study that can be exported for form a linear diagram. This can be done in the Results section, specifically under the Datasets subsection. Right clicking on the section header for subsets allows one to add, among several other types of objects, a **Cut Line 2D** object, which slices through the evaluated mesh to produce linear data. Two lines are evaluated in the scope of this study: one directly above the bottom PMTs, and one directly below the top PMTs. These lines pass directly above the elevated buffers which sit between the PMTs, rather than intersecting the geometry of the areas composed of PTFE. The top line runs from coordinates (0, 69.5) to (695, 69.5) in millimeters, and the bottom line runs from (0, -1543) to (695, -1543). These lines only have to be created once so long as changes to the COMSOL file are saved between runs.

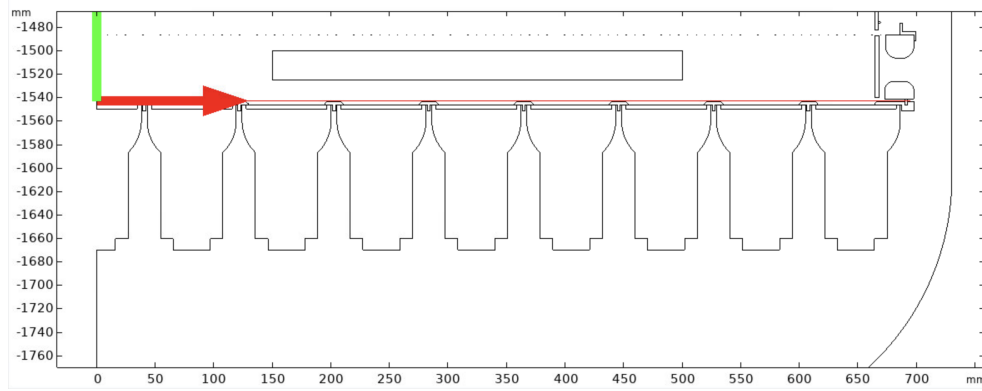


Figure 2.3: A diagram of the line of evaluation (red) near the bottom PMTs. This is a physical diagram depicting a where the 2D model is sliced to produce our line of evaluation. Note that the Red and Green arrows are the coordinates of the rendering and do not represent aspects of the physical geometry. Distances are noted in mm, with the left side of the detector being the center.

After the completion of these datasets, the final step for a simulation run is the exporting phase. Under the Export block of the Results section, one can export

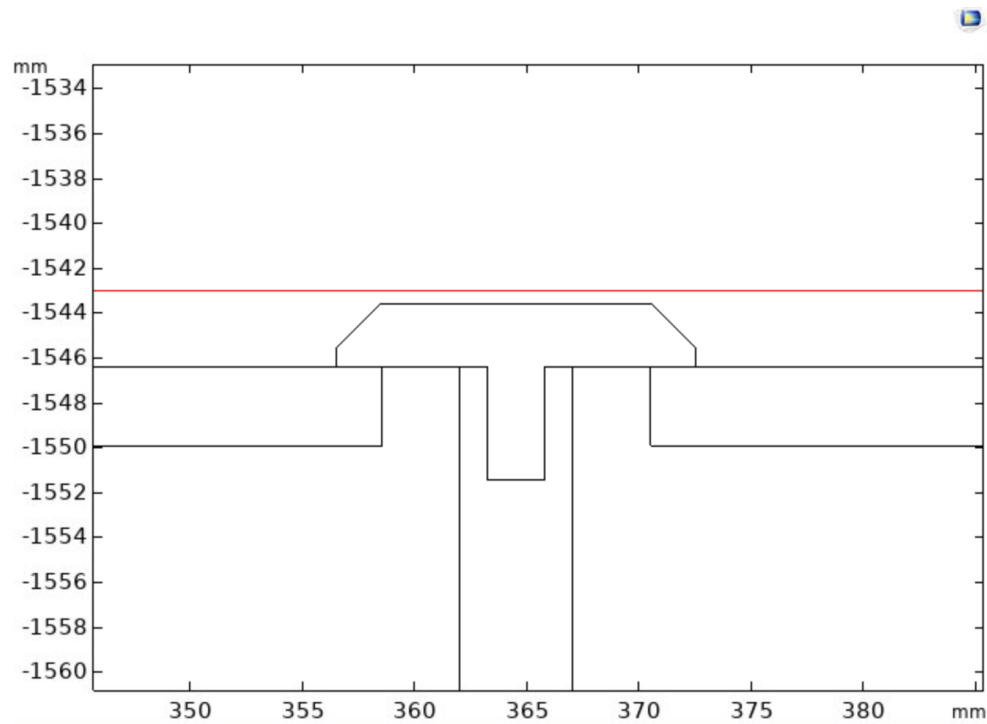


Figure 2.4: A close-up diagram of the line of evaluation (red) near the bottom PMTs, highlighting its separation from the PMT buffer. Distances are noted in mm.

datasets as files using the data export block. After creating this block, the dataset should be changed to the line upon which the analysis was performed. The expression block should be filled with the properties one wishes to export; in this case, the expression `es.normE`, with units V/m , is necessary. Positional data is exported by default. Within this block one can also choose the output type; we used the “Text” file option with the file extension `.csv` in the desired location. The next parameters are set as follows:

Points to evaluate in: From dataset

Data format: Spreadsheet

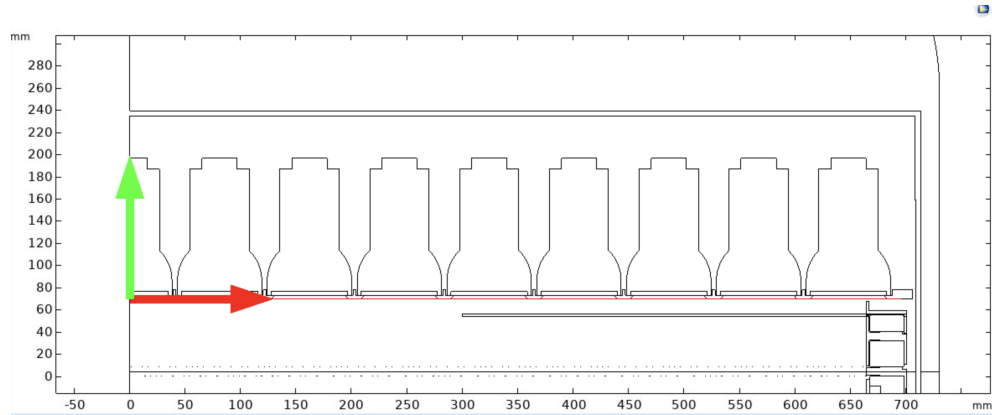


Figure 2.5: A diagram of the line of evaluation (red) near the top PMTs. This is a physical diagram depicting a where the 2D model is sliced to produce our line of evaluation. Note that the Red and Green arrows represent the coordinate system and do not represent aspects of the physical geometry. Distances are noted in mm, with the left side of the detector being the center.

Space Dimension: From dataset

Geometry Level: From dataset

Note that while COMSOL asks the user if they wish to overwrite a pre-existing file when using the browser functionality to choose a filepath, there is no such warning when hitting the Export button in the Results section. This means that if a filepath is typed into the “save as” box by hand, there will be no warning if data will be overwritten. Thus we have found it useful to change the filepath before running a new simulation, so as to minimize the chance of an accidental overwriting of data.

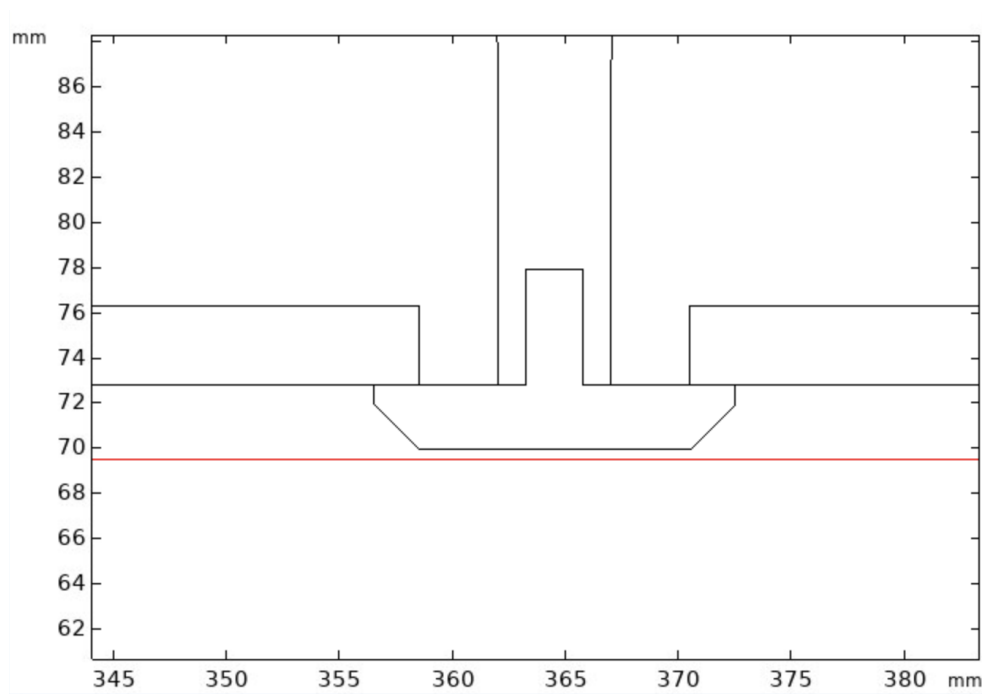


Figure 2.6: A close-up diagram of the line of evaluation (red) near the top PMTs, highlighting its separation from the PMT buffer. Distances are noted in mm.

Chapter 3

Methodology Utilizing Jupyter (Analysis Workflow)

After exporting the simulation data from COMSOL, the next phase of research consisted of python-based analysis. Using a Jupyter notebook [7], a directory was created that consisted of all datasets as well as the code itself. Modules Pandas, Numpy, and Matplotlib were utilized. Importing the data required reformatting the csv files as exported by COMSOL, removing the first 8 rows with the tag `skiprows=8` used during the input. These first eight rows are used as a header by COMSOL, describing simulation properties and units. After data was successfully imported, the first tests conducted were visual, plotting an example dataset based on radial position and calculated electric field norm to determine if results were as expected. The results initially appeared as seen in figure 3.1.

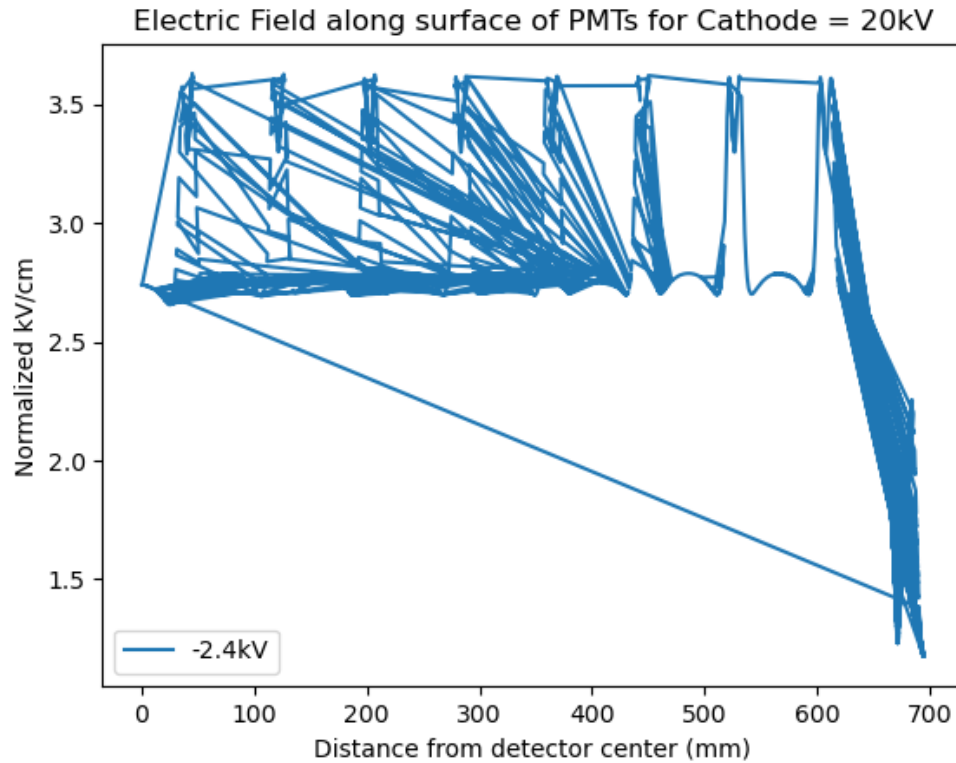


Figure 3.1: Initial view of lower PMT Electric Field Norms before data sorting. While a scatter plot would clean up these visuals, it would not resolve the underlying issue demonstrated here, which is that data points are not sequenced linearly.

While the data clearly reflects the graph seen in COMSOL, lines are seen crisscrossing the data. These are due to a non-deterministic ordering of data points by COMSOL during the exporting process; data is not returned in an ordered manner with regards to the innermost and outermost data points. After sorting the data, it can be processed by index.

In figure 3.2 we can observe the graph of the electric field norm along the lower PMTs with a Cathode voltage of -20kV and screening ring voltages of -2.4kV and -3.6kV, showing the range of forms seen for a given Cathode voltage. This is the

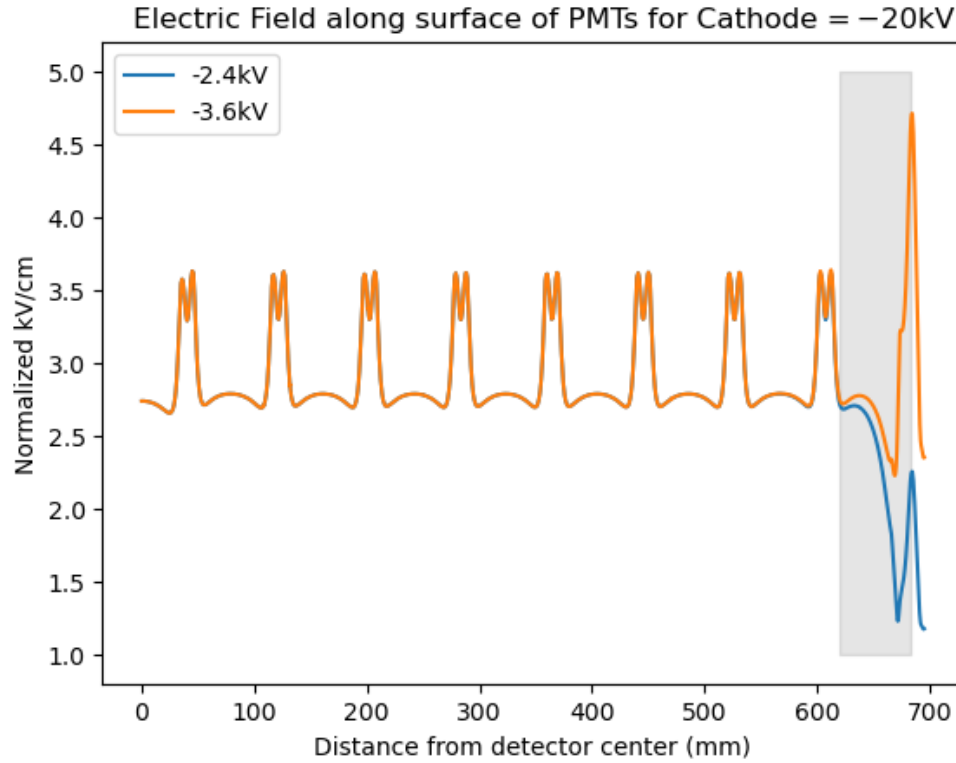


Figure 3.2: Example view of lower PMT Electric Field Norms after data sorting. Results with extreme voltages of the PMT are displayed to show the range of variance. Shaded region shows outer PMT.

approximate form of the electric field norm across the lower PMTs for all conducted runs, with a relatively constant field norm experienced across most PMTs, interrupted by the PMT buffers. The outermost PMT, highlighted in grey, has the most extreme distribution of electric field norms. The upper PMTs show a similar pattern, with an example at Anode voltage 2kV, gate voltage at -6kV, and screening ring voltages of -0.4kV and -1.2kV shown in figure 3.3. These patterns are very similar, with the most significant differences being the electric field norms across the outermost PMT.

As outlined in the introduction, our target was to evaluate the RMS (root mean

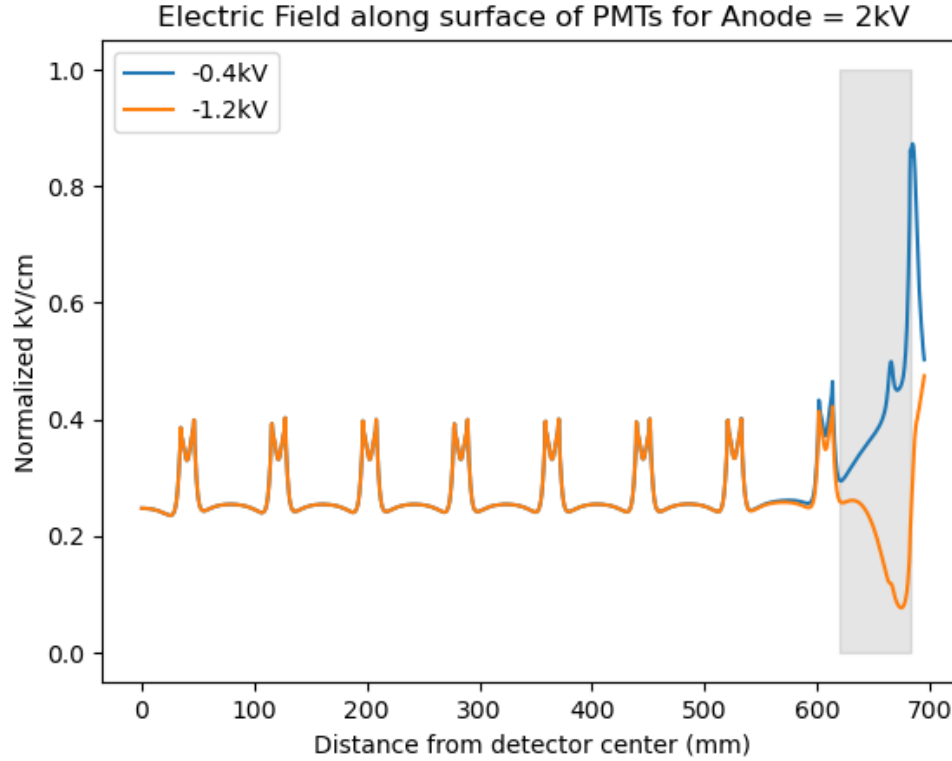


Figure 3.3: Example view of upper PMT Electric Field Norms after data sorting. Results with extreme voltages of the PMT are displayed to show the range of variance. Shaded region shows outer PMT.

square) of the electric field on the PMTs. In minimizing the RMS, we minimized the deviation of the voltages experienced from the mean. As the outer PMT has by far the greatest variance among the PMTs, only this PMT was analyzed for RMS. The second-outermost PMT on the upper PMTs also experiences a visible change in electric field between the highest and lowest PMT voltage, though it has a much smaller variance than the outer PMT, and thus a negligible impact. In order to calculate the RMS for each run, the following procedure was applied: First, after sorting by radius, two regions are defined, one including all points within the outermost PMT and one including all points within all other PMTs. The points within all other PMTs are

averaged to determine the mean electric field norm for the dataset, with this used as the mean in our RMS under the name `norm`. It is important to note that this norm is calculated only using the data points within the inner PMTs themselves, and not the electric field norms seen over the buffers between PMTs. This is so that a consistent benchmark could be used for the mean electric field norm between different TPC voltages. The electric field on these buffers is visible in diagrams as evenly spaced spikes above the normal electric field norms of the PMTs. The data mask is performed using a function which defines the regions contained within each of the inner PMTs and a function which checks each points radius to determine if they should be added to a list of valid data points based on data array index. The same method is used to define points contained within the outer PMT. The RMS is then calculated using the following formula:

$$\sqrt{\frac{1}{N} \sum_{i=1}^N (E_i - \bar{E})^2}$$

Which is written in python as:

```
np.sqrt((1/len(rms_mask)) * np.sum(((rms_mask['es.normE (V/m)']) - norm)**2 ))
```

With `np.sqrt` representing the square root function, `rms_mask` being the set of results on the region of the detector for which the value is calculated, `len(rms_mask)` being the number of data points in the result section, `rms_mask['es.normE (V/m)']` being the data points for the electric field norm within the region. After this is performed, the RMS is acquired for every combination of Anode/Cathode voltage and screening ring voltage. We can graph the RMS values based on the voltage of the screening ring to acquire a graph as seen in figure 3.4 which depicts the data taken for the Anode

with a voltage of 2kV. This data was found to form a hyperbolic correlation for each evaluated Anode/Cathode voltage. The hyperbolic formula used here was defined as:

$$\sqrt{|b^2(1 + \frac{(x - x_0)^2}{a^2})|} + y_0$$

b and a are shape parameters and x_0 and y_0 represent the location of the focus, with x_0 also being the x-location of the minimum. This formula is used to evaluate each of the hyperbolas created to extrapolate from the evaluated data points. The python function `curve_fit` [8], part of the `scipy.optimize` module, calculates the optimal curve via the least squares method; this was used to determine the correlation and uncertainty, in the form of covariance, for each hyperbola. The minimum points of each hyperbola corresponded to the voltage of the screening ring that would produce the smallest RMS across the outer PMT. These voltages were then plotted alongside Anode/Cathode voltage to produce an equation defining the relationship between Anode/Cathode voltage and ideal screening ring voltage.

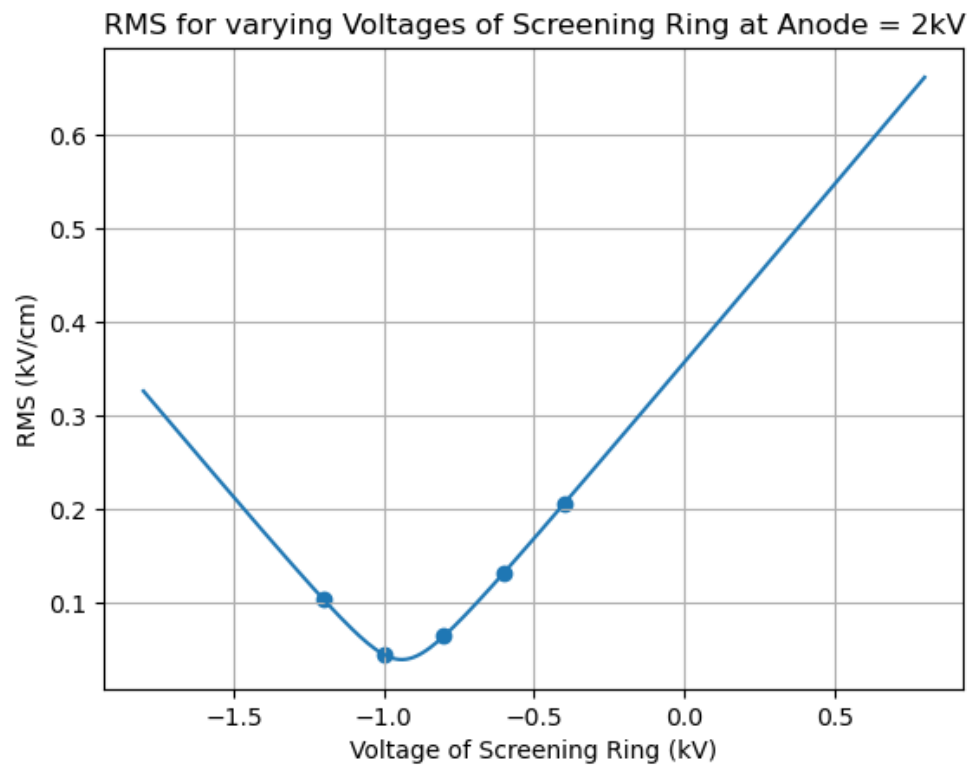


Figure 3.4: Data points for RMS based on TSR voltage for an Anode with voltage 2kV. A hyperbola defining the relationship between the TSR voltage and RMS is shown.

Chapter 4

Results

Data collection for this study was performed in three primary segments: one determining the relationship between the Cathode and BSR, one between the Anode and TSR, and one determining the effect of the Screening Meshes as opposed to Screening Rings on the electric field norm along the surface of the PMTs. Beginning with the Cathode data, five values for the Cathode voltage were used, each with seven data points for different BSR voltages. One exception was made for a Cathode voltage of 10kV, for which the BSR voltage of -1.4kV was evaluated; this BSR voltage was later found to be too similar in voltage to the PMTs (-1.3 kV). When a Screening Ring is of similar voltage to the PMT, it was found that data began to differ from the hyperbolic correlation which dominated other data elements, with trends present in other data reversing. Furthermore, data points in the section of parameter space with similar PMT and screening ring voltages ($\Delta < 0.2\text{kV}$) featured high RMS values which made them both unfeasible for use within XENONnT and useless for determining trends

Cathode	BSR						
-10kV	n/a	-1.6kV	-1.8kV	-2.0kV	-2.2kV	-2.4kV	-2.6kV
-12.5kV	-1.8kV	-2.0kV	-2.2kV	-2.4kV	-2.6kV	-2.8kV	-3.0kV
-15kV	-2.0kV	2.2kV	2.4kV	2.6kV	2.8kV	3.0kV	3.2kV
-17.5kV	2.2kV	2.4kV	2.6kV	2.8kV	3.0kV	3.2kV	3.4kV
-20kV	2.4kV	2.6kV	2.8kV	3.0kV	3.2kV	3.4kV	3.6kV

Table 4.1: The voltage combinations involving the Cathode and BSR for which data was evaluated.

within the critical data collection region. Table 4.1 lists the combinations evaluated.

After sorting the data as described in the Methodology Utilizing Jupiter section, the test runs for a Cathode of -10kV are shown in figure 4.1. Note the misalignment of the trend observed in all other test cases within the critical region for an evaluation with a BSR voltage of -1.4kV. While this dataset was calculated, this reversal of trends meant that it was not used in later evaluation and curve fitting. As well, figure 4.2 displays all Cathode data. Note that the alignment of datasets seen on the left and center of the diagram are the result of the consistent Cathode voltages, while the diverging values on the diagram's right within the critical region are the result of the varying values of the BSR.

After all data was collected, each set of data sharing a value for the Cathode was evaluated using the RMS evaluation outlined in the Methodology Utilizing Jupyter section. Through this method, the RMS for every Cathode/BSR combination was calculated. A hyperbola of best fit was calculated for each Cathode value, extrapolating from the data points to allow the determination of a BSR voltage that would result in minimum RMS over the outer PMT. The parameters of each of these hyperbolas

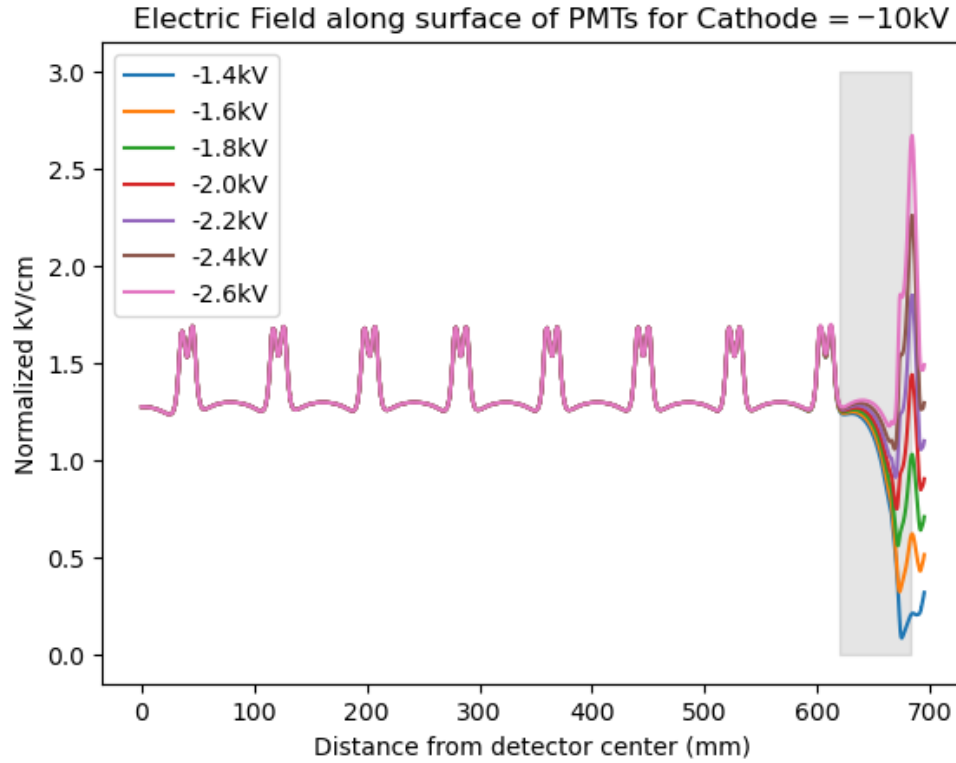


Figure 4.1: A view of all Electric Field Norm data sets for a Cathode voltage of -10kV. The later unused dataset for BSR voltage of -1.4kV is shown in blue. Note how the local maximum at approximately 700mm from the center of the detector is far smaller in this datapoint. The critical region of the outer PMT is once again highlighted in grey.

is shown in table 4.2, along with figure 4.3, a chart displaying the five hyperbolas evaluated for the Cathode.

After the evaluation of each hyperbola, their minima were determined – with each hyperbola’s minimum’s x -value equal to the value x_0 used in their definition. These minima define the ideal values of the BSR’s voltage for a given voltage of the Cathode. These ideal values, in turn, define an equation connecting any specific Cathode voltage to the BSR voltage which minimizes the RMS on the outer PMT.

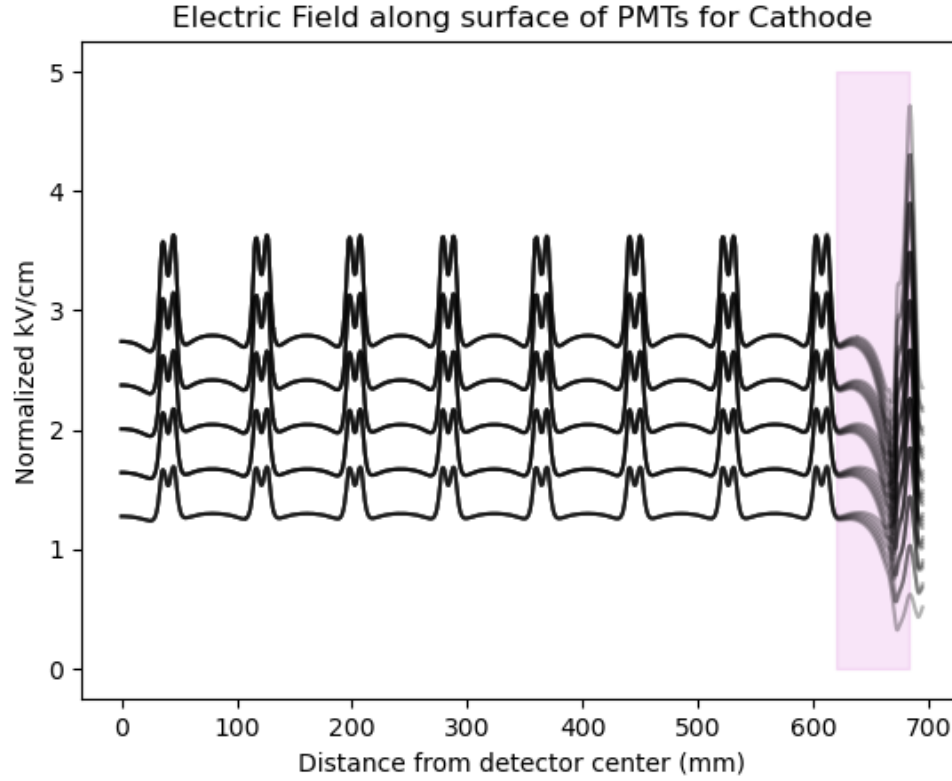


Figure 4.2: A view of all Electric Field Norm data sets for all Cathode voltages evaluated, with lines made semi-transparent to allow the distribution of values to be evident. The critical region of the outer PMT is highlighted in pink.

Using a linear equation $y = mx + b$ and the function `curve_fit` described earlier, this equation is determined to be approximately linear, with the final equation (with uncertainty) as follows:

$$BS = (0.1117 \pm 0.0001) * C - (1.149 \pm 0.002)$$

The reason this result is so linear can be traced back to the origins of electric field norm as a gradient, $E_n = |E| = |\nabla V|$. The field norm is linearly proportional to

Cathode	x_0	y_0	a	b
-10kV	-2.2665 ± 0.0009	6000 ± 1000	0.14 ± 0.02	13000 ± 2000
-12.5kV	-2.5432 ± 0.0007	6000 ± 2000	0.20 ± 0.02	18000 ± 2000
-15kV	-2.824 ± 0.001	5400 ± 800	0.263 ± 0.007	24500 ± 800
-17.5kV	-3.103 ± 0.002	5200 ± 600	0.321 ± 0.005	-30300 ± 600
-20kV	-3.382 ± 0.002	5200 ± 700	-0.377 ± 0.006	35700 ± 700

Table 4.2: The equations for the hyperbolas which define the relationship between the voltage of the BSR and the RMS of the outer PMT for each Cathode voltage evaluated.

the change in Voltage over a set distance. Thus a linear change in the voltage of the Cathode resulting in an approximately linear change in the electric field norm, and in the ideal voltage of the BSR, is to be expected.

A similar methodology was employed to determine the correlation between the Anode voltage and the ideal TSR voltage. Before data was collected, it was questioned if the value of the Gate, positioned below the Anode, would have a significant impact on the electric field norms experienced by the PMTs. With a much closer distance than that between the Gate and Cathode, it was worth determining if the voltage of the Gate would have an impact. Electric Field Norms are the magnitude of the slope of the Voltage, so changing the voltage of a component that is not entirely shielded from the PMTs could have an impact on the field norm experienced at the PMT surface.

While the XENONnT collaboration plans on grounding the Gate at zero volts, there was deliberation about whether it may be necessary to adjust its voltage to minimize the Electric Field Norm. Tests were thus done to assess the Gate's impact on the PMT's. Two data sets were taken assuming an Anode voltage of 6kV and

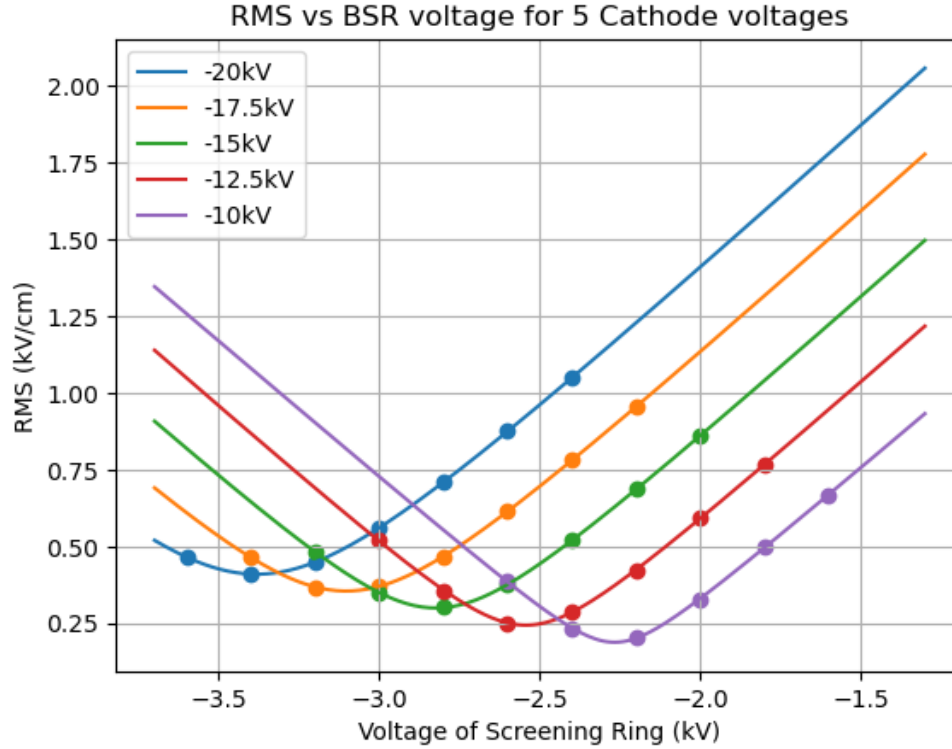


Figure 4.3: Data points for RMS based on TSR voltage for 5 Cathode voltages. A Hyperbola defining the relationship between the TSR voltage and RMS for each Cathode voltage is shown.

a TSR voltage of 0.4kV. In one dataset a value of -2kV was used for the Gate, while the other saw a Gate grounded at 0kV. It was determined that the two data sets had noticeable differences; while the critical region in the outer PMT did not possess very significant changes between runs, the other PMTs had values differing by approximately 0.05 kV. This was deemed insignificant as the variance was well below the experimental uncertainty in voltages produced within the XENONnT detector. However, as a precaution, it was determined that a consistent voltage difference of 8kV between the Gate and Anode should be possessed in all data analyzed in order to minimize the effect of the Gate on data. Thus it is worth noting that a different Gate

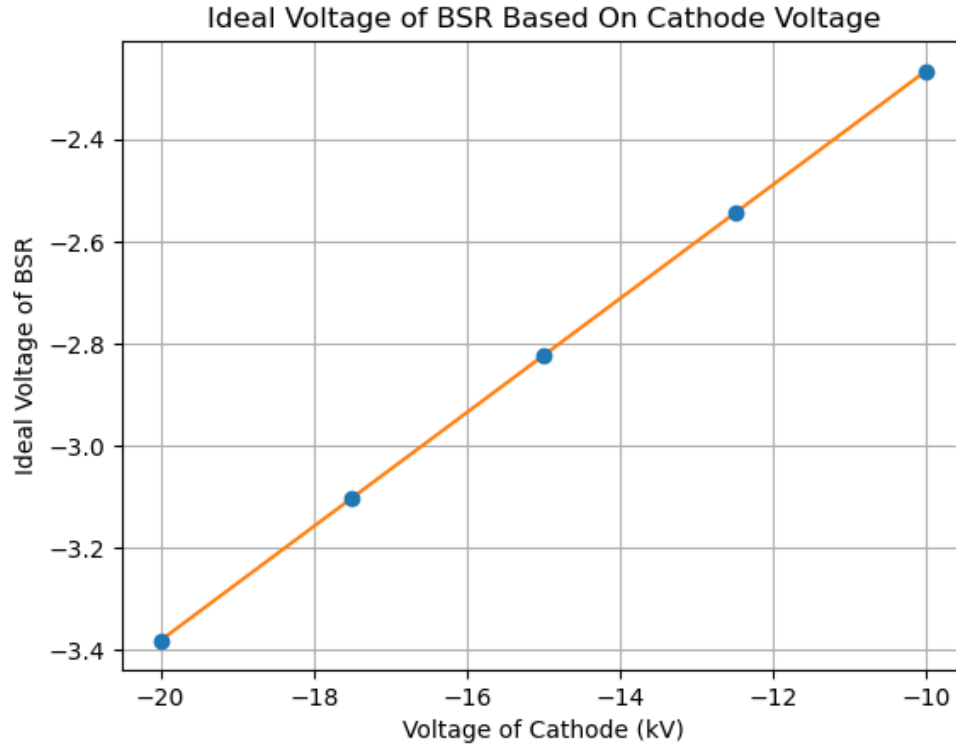


Figure 4.4: The line of best fit correlating the Cathode voltage to the BSR voltage which minimizes the RMS on the outer PMT.

value will lead to a different absolute electric field if the simulation was repeated. The results of the Gate voltage are discussed further in the conclusions section of this report.

The data evaluated for the Anode is less consistently sized; due to the lower voltage difference between the Anode and PMTs, less data was collected for the TSR with voltages closer to those of the PMTs. This was because Anode values close to the PMT voltage resulted in a smaller critical region. Runs for four voltages of the Anode were calculated; originally there were plans to sweep the parameter space for an Anode with 0kV voltage, but it was found that the closer charge of the Anode

Anode	Gate	TSR							
2kV	-6kV	-1.2kV	-1.0kV	-0.8kV	-0.6kV	-0.4kV			
4kV	-4kV	-1.2kV	-1.0kV	-0.8kV	-0.6kV	-0.4kV	-0.2kV	0.0kV	
6kV	-2kV	-1.2kV	-1.0kV	-0.8kV	-0.6kV	-0.4kV	-0.2kV	0.0kV	0.2kV 0.4kV
8kV	0kV	-0.6kV	-0.4kV	-0.2kV	0.0kV	0.2kV	0.4kV	0.6kV	0.8 kV

Table 4.3: The voltage combinations involving the Anode and TSR for which data was evaluated.

Anode	x_0	y_0	a	b
2kV	-0.9402 ± 0.0006	-600 ± 200	-0.117 ± 0.004	4500 ± 200
4kV	-0.461 ± 0.001	900 ± 300	-0.149 ± 0.009	5400 ± 400
6kV	0.019 ± 0.002	1100 ± 200	0.221 ± 0.007	8000 ± 300
8kV	0.4961 ± 0.0003	530 ± 70	0.311 ± 0.002	11490 ± 80

Table 4.4: The equations for the hyperbolas which define the relationship between the voltage of the TSR and the RMS of the outer PMT for each Anode voltage evaluated.

to the PMTs, which forced a screening ring of similar voltage, made this section of parameter space irrelevant. The relevant parameter space is between the bounding voltage values.

After data for the Anode and TSR was collected, a similar analysis was conducted to that described for the Cathode and BSR. All data is shown in table 4.3, followed by the hyperbolic curves resulting from RMS analysis in figure 4.6. The upper and lower PMTs are aligned with one another, and thus the same critical region occurred in both datasets. Note that the maximum electric field norm across the upper PMTs is less than that experienced by the bottom PMTs. The hyperbolas were then calculated according to the same process used for the lower PMTs, with the parameters of the four hyperbolas defined outlined in table 4.4.

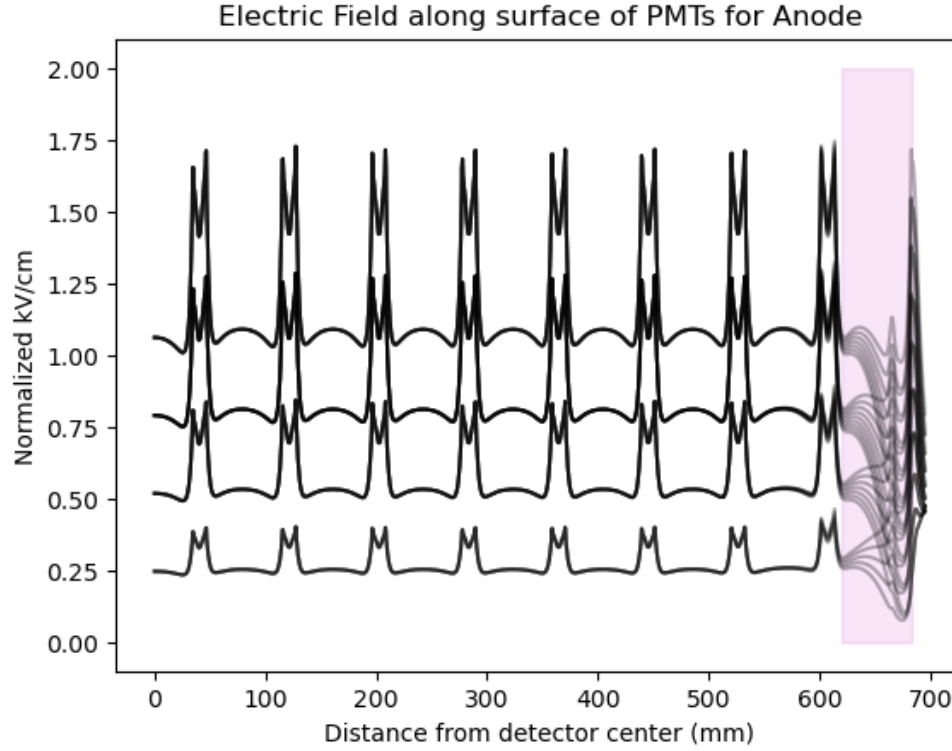


Figure 4.5: A view of all Electric Field Norm data sets for all Anode voltages evaluated, with lines made semi-transparent to allow the distribution of values to be evident. The critical region of the outer PMT is highlighted in pink.

The minimum of each hyperbola, corresponding with parameter x_0 , were again used to determine a linear equation, this one determining the relationship between Anode voltage and ideal TSR voltage. The final equation (with uncertainty) is as follows:

$$TS = (0.2394 \pm 0.0001) * A - (1.4189 \pm 0.0007)$$

Lastly, we examined what would occur if the TSR and BSR were to be replaced with wire arrays. This was the approach taken by XENONnT previously, though this system is now under reconsideration with the adoption of the screening ring as a less

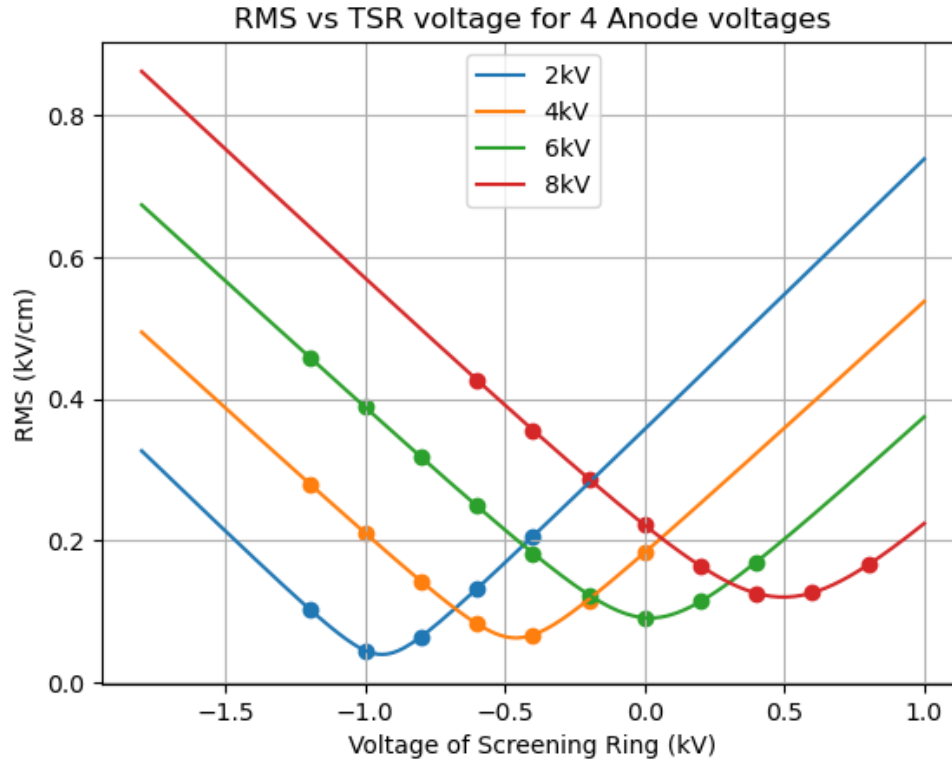


Figure 4.6: Data points for RMS based on TSR voltage for 4 Anode voltages. A hyperbola defining the relationship between the BSR voltage and RMS for each Anode voltage is shown.

invasive method towards the minimization of the RMS along the lower and upper PMTs. We wished to compare the electric field norm for the PMTs with the norm of the same PMTs with the top and bottom wire arrays discussed earlier. For this step the wires generated for an earlier iteration of the model, which were deactivated for the main body of research, were reactivated. The wire arrays were set to -1.3 kV in each test, the same as the PMTs that they were intended to protect. Tests on the Cathode/bottom PMTs and Anode/top PMTs were conducted at the same time, as the voltage of the Cathode and Anode were assumed to have no effect on one another, and this allowed for higher compute time efficiency. Note that for these

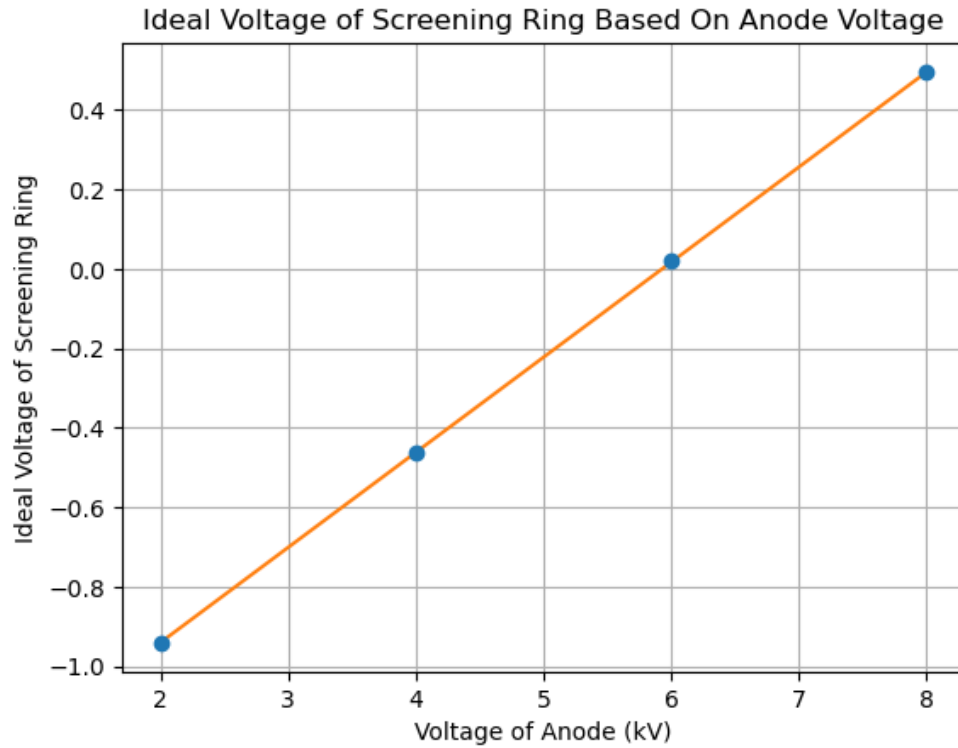


Figure 4.7: The line of best fit correlating the Anode voltage to the TSR voltage which minimizes the RMS on the outer PMT.

tests the Gate was set to a consistent value of 0kV, as opposed to the previous Anode tests in which the voltage of the Gate was 8kV less than the voltage of the Anode.

The distribution of electric field norms for the Anode without the wire mesh was found as a point of comparison. The TSR had little effect on these values, and thus different runs were averaged. The relationship between the voltage of the Anode in kV and the mean electric field norm of the PMTs in kV/cm was found by using a mask to select the PMTs and taking the mean of the selected data. The relationship

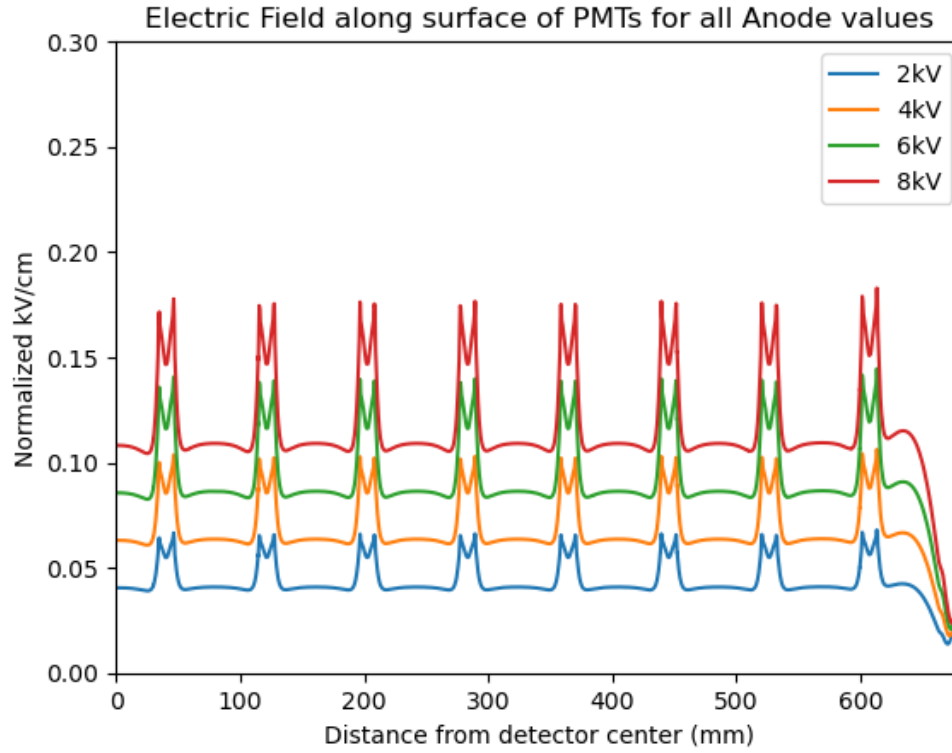


Figure 4.8: A view of the relationship between Anode voltage and Electric field norm when a screen of wires is used in place of the TSR. Four potential Anode voltages are described, with field norms being lowered when compared to those experienced when a TSR is used.

was found to be as follows:

$$AnodeFieldNorm = (0.13711 \pm 0.00002) * A - (0.0234 \pm 0.0001)$$

The distribution of electric field norms for the Anode with the wire mesh resembles those for the Anode without this mesh, with the primary point of variance being magnitude.

$$AnodeFieldNormWires = (0.01134 \pm 0.00002) * A + (0.01801 \pm 0.00007)$$

The field norm of the bottom PMTs, by contrast, had high variance, as seen in figures 4.10 and 4.11. This can be attributed to two primary causes: firstly, the bottom wires are much closer to their PMTs than in the case of the top screening wires, which means that the separation between wires is not at such a distance where it becomes negligible. Secondly, the line of evaluation, which is elevated above the surface of the PMTs, is proportionately much closer to the wires than in the case of the upper wires. The line of evaluation is always the same distance from the PMTs, but since the distance to the screening mesh is smaller, the influence of each individual wire becomes greater. These two factors lead to a high observed variation along the line of evaluation. The actual variance in electric field norm experienced by the PMTs is likely non-negligible, but still not as dramatic as the phenomenon evaluated at this line. The mean field norms for the Cathode are also weaker when compared to their values with the BSR. The equation found for the mean field norm across the PMTs for the Cathode (both with and without screening wires) in kV/cm is as follows:

$$CathodeFieldNorm = (0.148048 \pm 0.000007) * C - (0.1927 \pm 0.0001)$$

$$CathodeFieldNormWires = (0.0482 \pm 0.0004) * C - (0.063 \pm 0.006)$$

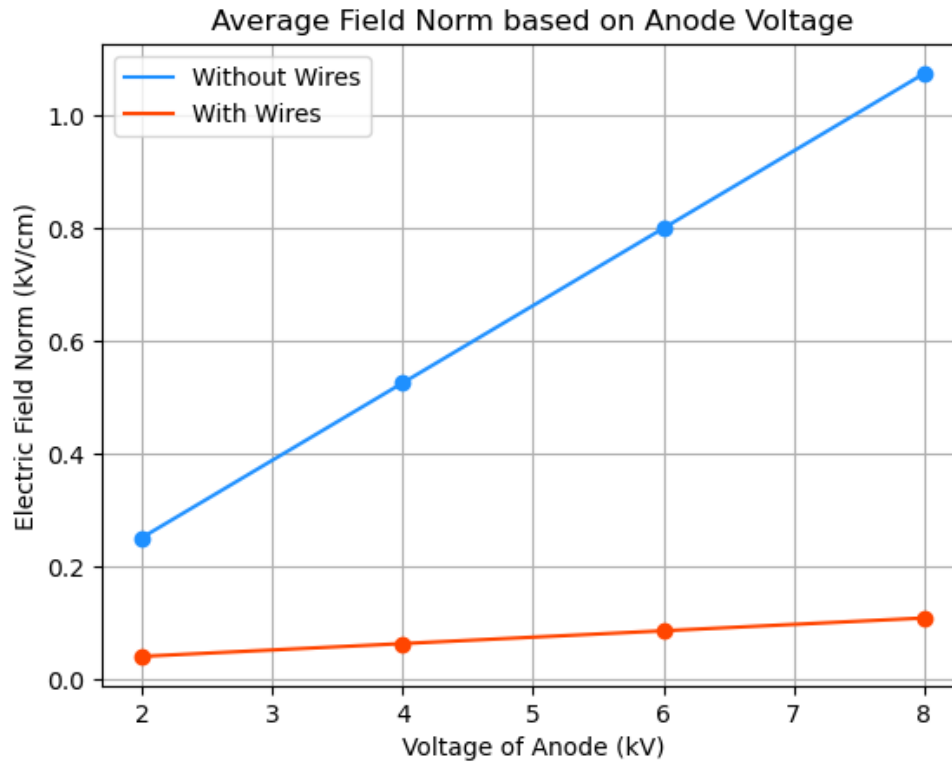


Figure 4.9: The relationship between Anode Voltage and electric field norm can be represented as a linear equation. This relationship is compared between the XENONnT model with or without the wire mesh. Note that the Gate was set to a voltage 8kV below that of the Anode in the runs utilizing the model without wires, while it was always set to 0kV when the wire mesh was used. It is believed that with 2 sets of wire meshes between the Gate and PMTs, the impact of the Gate on the Top PMT field norm would be far smaller when using a wire mesh.

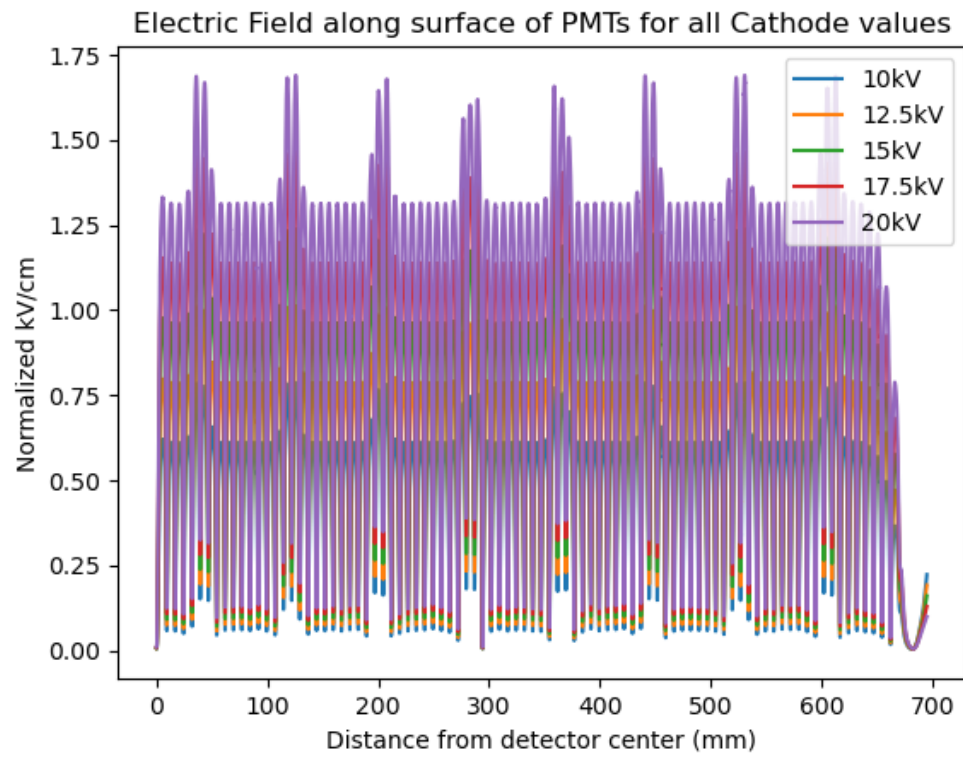


Figure 4.10: A view of the relationship between Cathode voltage and Electric field norm when a screen of wires is used in place of the TSR. Five potential Cathode voltages are described. Despite high variance, field norms are lowered when compared to those experienced when a TSR is used.

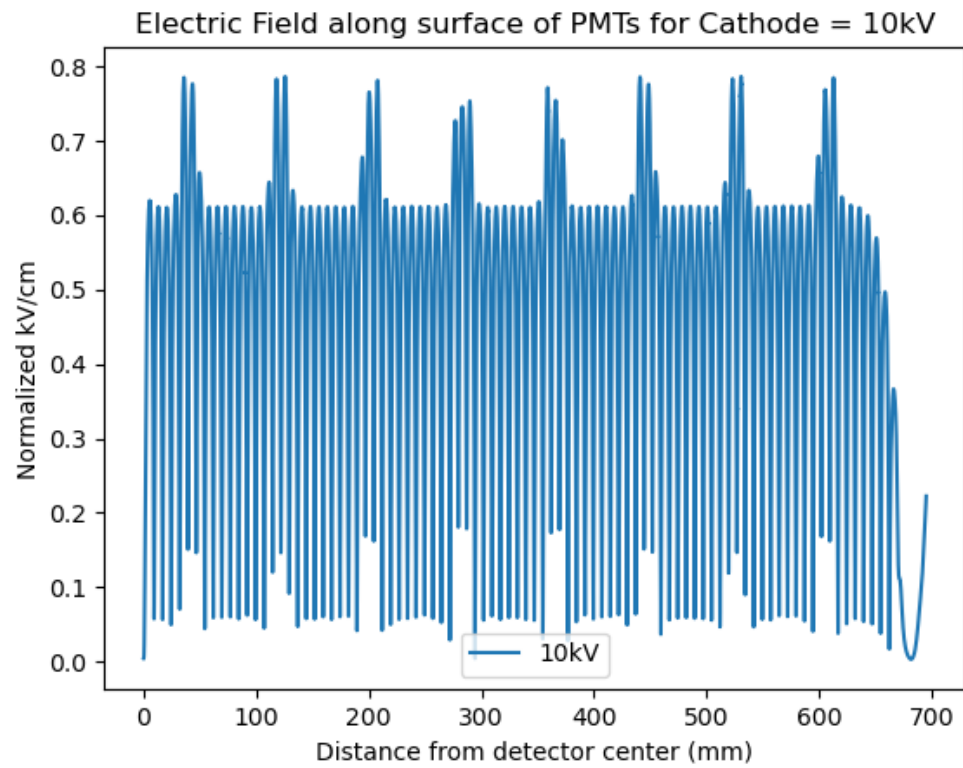


Figure 4.11: A view of the relationship between Cathode voltage and Electric field norm when a screen of wires is used in place of the TSR. One diagram is shown for the sake of clarity.

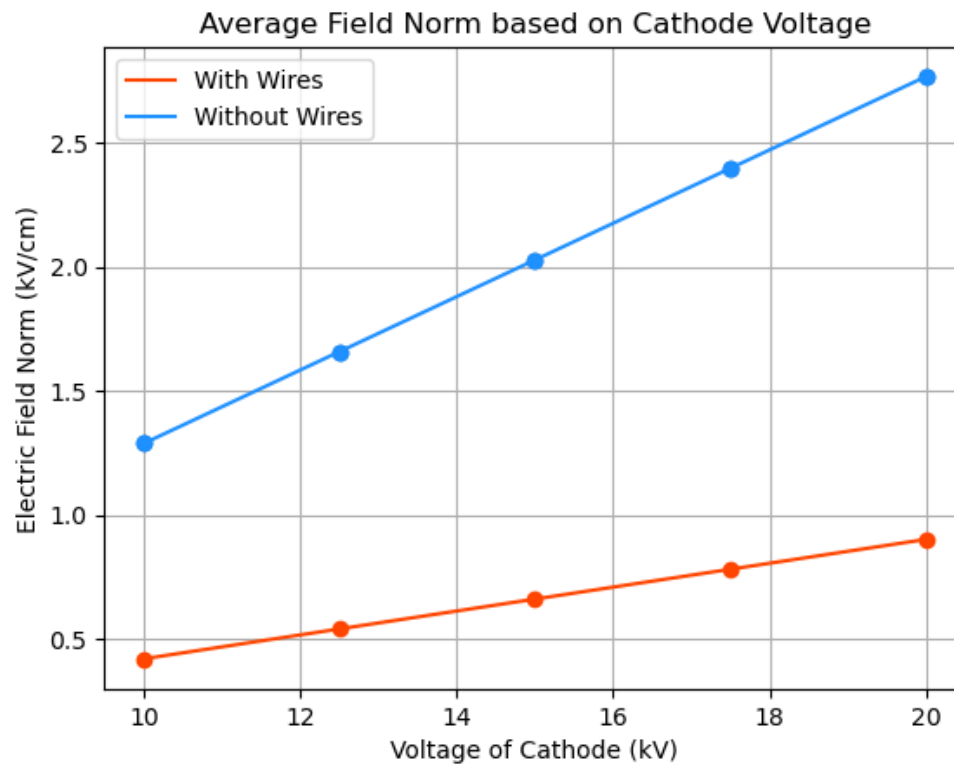


Figure 4.12: The relationship between Cathode Voltage and electric field norm when wires are used in place of the BSR can be represented as a linear equation. It is compared to the XENONnT model without a wire mesh.

Chapter 5

Conclusion

In summary, decisive relationships have been found between the Anode and Cathode voltages and both the ideal voltages of field shaping rings, and between the voltages and their mean electric field norms. The primary application of this analysis is the creation of these equations, which govern the relationships between voltages and the conditions experienced by the PMTs. These relationships will allow the members of the XENONnT collaboration to better understand the ideal voltages of the screening rings for given Anode and Cathode voltages. In future uses of the detector these equations can be used to determine the voltages assigned.

As a sanity check of the equations derived, we determined what the TSR/BSR would be if the Anode/Cathode are set to -1.3kV , the same voltage as the PMTs. It is expected that under these conditions the TSR and BSR should also be set to -1.3kV , as the screening rings do not have to accommodate for any electric fields. If

the Cathode is set to a value of -1.3kV , then the equations imply the BSR should also be set to -1.3kV , which matches with expected behavior. However, for an Anode of -1.3kV , it implied that a more negative TRS voltage is needed. This is likely due to the voltage of the Gate, as the Gate is not neutrally charged. The Anode equation will change the calculations of ideal voltage assumed up until this point, which is one of the most direct applications of this study. This variation is best explained by the non-local effects of other charged components, especially the Gate. Our simulations assumed a constant 8kV difference between the Anode and Gate, with the leakage of the Gate affecting the region around the TSR. This might indicate a systematic uncertainty to how the simulations represent reality.

Further continuations of this research could seek to expand upon the findings of this paper to create a more complete understanding of the variables whose relationships were sought in this study. For example, analyzing the higher-dimensional relationship between the Gate, Anode, and TSR may be of use. The Gate was seen to have an effect on the top PMTs, meaning this relationship may be of interest. As well, a further analysis comparing different types of screening systems may be of use. The axisymmetric model of the detector lends itself well to models of symmetrical structures such as rings, but the non-rotationally symmetric nature of a series of parallel wires makes them difficult or impossible to model under these circumstances. Due to the nature of an axisymmetric simulation the wires were approximated by a series of concentric rings for the purposes of this study. While this is equivalent to a cross section perpendicular to the wires, different properties might be observed on different cross sections of the detector. Using a wire mesh or a more complicated grid may

produce more accurate results, though this would inherently be 3-dimensional. 3D models are highly computationally expensive, so other approximations may need to be used in these cases. Lastly, the axisymmetric view also assumes a fixed distance of the outer PMT, though the true outer PMTs vary in distance. This is due to a hexagonal lattice structure used in the arrangement of PMTs, visible in figure 1.4. Testing with “slices” of the 3D model which showcase other combinations of PMTs could allow XENONnT to determine the correlations described here to greater fidelity.

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